STAT/MA 41600
In-Class Problem Set #17: September 28, 2026
Solutions by Mark Daniel Ward

Problem Set 17 Answers

1. Among the first r rolls, we want either 0 or 1 values of "3". (Otherwise, if there are 2 or more values of "3", then we will have $X \leq r$.) It follows that $P(X > r) = (5/6)^r + \binom{r}{1}(5/6)^{r-1}(1/6)$.

2a. No, W is not a Negative Binomial random variable because X and Y are associated with different probabilities of success on each trial.

If W was a Negative Binomial random variable, then we need $r = 2$ since $W \geq 2$. We would also need to have $P(W = 2) = P(X = 1 \ \& \ Y = 1) = (2/5)(5/9) = 2/9 = p^2$, so we would need $p = \sqrt{2/9}$, which seems strange. In particular, this would imply that $P(W = 3) = \binom{3}{2}(1 - \sqrt{2/9})(\sqrt{2/9})^2$. In contrast, we have $P(W = 3) = P(X = 2 \ \& \ Y = 1) + P(X = 1 \ \& \ Y = 2) = (3/5)(2/5)(5/9) + (2/5)(4/9)(5/9) \neq \binom{3}{2}(1 - \sqrt{2/9})(\sqrt{2/9})^2$. So we conclude that W cannot be a Negative Binomial random variable.

2b. No, V is not a Negative Binomial random variable because V takes on the possible values 10, 20, 30, ..., but Negative Binomial random variables take on values of the form $r, r+1, r+2, \ldots$.

3. The expected number of rolls altogether is $r/p = 9/(1/6) = 54$. Among these rolls, exactly 9 of them are red, so the expected number of non-red rolls is $54 - 9 = 45$. Each of those non-red rolls has probability $2/5$ of being blue and probability $3/5$ of being yellow. Thus, the expected number of blue rolls is $(45)(2/5) = 18$.

4a. We have $E(X + Y) = E(X) + E(Y) = 1/(2/5) + 1/(5/9) = 5/2 + 9/5 = 43/10$.

4b. We have $E(X - Y) = E(X) - E(Y) = 1/(2/5) - 1/(5/9) = 5/2 - 9/5 = 7/10$.

4c. Since X and Y are independent, we have $Var(X + Y) = Var(X) + Var(Y) = (3/5)/(2/5)^2 + (4/9)/(5/9)^2 = 519/100 = 5.19$.

4d. Since X and Y are independent, we have $Var(X - Y) = Var(X) + (-1)^2 Var(Y) = 519/100 = 5.19$.

5a. Yes, V is the sum of 3 independent Geometric random variables with the same probabilities of successes on each of their trials (namely, p), so V is a Negative Binomial random variable with parameter p and parameter $r = 3$.

5b. We have $P(V > 10) = \binom{10}{0}(1-p)^{10} + \binom{10}{1}(1-p)^9 p + \binom{10}{2}(1-p)^8 p^2$.

5c. We have $P(V > k) = \binom{k}{0}(1-p)^k + \binom{k}{1}(1-p)^{k-1} p + \binom{k}{2}(1-p)^{k-2} p^2$.